Comparative Benchmark of State-of-Charge Observers and Physics-Informed Neural Networks Across Battery Aging and Thermal Regimes

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*While battery State-of-Charge (SOC) estimation has been extensively studied, existing comparative benchmarks predominantly evaluate fresh cells at room temperature, obscuring algorithmic vulnerabilities under operational aging and thermal extremes. This study establishes a rigorous multi-attribute benchmark evaluating representative nonlinear observers, deep recurrent networks, and physics-informed models (EKF, AEKF, DEKF, SMO, BiLSTM-Attn, and PINN-SOC) across standardized driving cycles. Algorithms are systematically benchmarked across progressive degradation (*100% → 80% *SOH), wide thermal regimes (*0°C *to* 45°C*), flat-plateau* $LiFePO_4$ *chemistry, and series 8S pack inconsistency. Findings reveal that while uncompensated observers suffer catastrophic 45-fold accuracy decay as internal impedance grows, parameter-tracking DEKF confines lifespan RMSE to* 0.585%*. Furthermore, we propose a lightweight hybrid Physics-Informed Dual Kalman Filter (PIDKF) that achieves* 0.414% *RMSE on degraded NMC and* 0.149% *on degraded LFP cells. Finally, cycle-accurate profiling on ARM Cortex-M microcontrollers maps the empirical Pareto efficiency frontier, providing actionable selection guidelines for industrial battery management systems.*

Keywords: State-of-charge, Lithium-ion battery, Battery degradation, Dual Extended Kalman filter, Physics-informed neural network, Electrochemical impedance spectroscopy, Embedded BMS, ASME JEECS

Nomenclature

$\mathbf{A}_k, \mathbf{B}_k, \mathbf{g}_k$ = discrete state transition, input, and hysteresis vectors
C_1, C_2, C_{dl} = polarization and double-layer capacitances (F)
$\mathbf{C}_{\theta,k}, \mathbf{H}_k$ = parameter sensitivity and observation Jacobian matrices
E_a, R = Arrhenius activation energy ($\text{J} \cdot \text{mol}^{-1}$) and gas constant
$I_k, I(t)$ = cell load current at time step k, positive for discharge (A)
$\mathbf{K}_k, \mathbf{L}$ = Kalman filter gain and sliding mode feedback gain
$M(z), V_h$ = maximum dynamic hysteresis function and hysteresis state (V)
$\mathbf{P}_k, \mathbf{Q}_k, \mathbf{R}_k$ = state error, process, and measurement covariance matrices
$Q_{\text{act}}, Q_{\text{nom}}$ = actual degraded and nominal pristine capacities ($\text{A} \cdot \text{h}$)
R_0, R_1, R_2 = bulk ohmic and dynamic polarization resistances (Ω)
$R_{\text{ct}}, R_{\text{sei}}$ = charge-transfer and SEI layer resistances from EIS (Ω)
$S, \mathbf{x}_k, \mathbf{y}_k$ = sliding manifold, state vector, and observation variable
T, T_{amb} = battery core temperature and ambient temperature (K or °C)
$U_1, U_2, \delta V_{\text{pol}}$ = RC overpotentials and unmodeled residual compensation (V)
V_t, V_k, V_{ocv} = terminal and thermodynamic open-circuit voltages (V)
$\mathbf{w}_k, v_k$ = process and measurement noise signals
$Z(j\omega), Z_w$ = electrochemical impedance and Warburg diffusion impedance (Ω)

Greek Letters

α, σ_w = constant-phase element exponent and Warburg coefficient
$\beta, \lambda_{\text{SH}}$ = hysteresis decay rate and Sage-Husa forgetting factor
$\gamma_{\min}, \eta_s, \Phi$ = sliding mode injection gain, margin, and disturbance bound
$\Delta t, \mu$ = discrete sampling interval (1.0 s) and boundary thickness (V)
η, κ = Faradaic coulombic efficiency and condition number
κ_0, κ_1 = Butler-Volmer overpotential bound coefficients
τ_1, τ_2 = electrochemical polarization time constants ($\tau_i = R_i C_i$, s)
$\boldsymbol{\theta}_k$ = parameter state vector in Dual Kalman Filter $[R_{0,k}, 1/Q_{\text{act},k}]^T$

Subscripts and Superscripts

act, nom, eff = actual degraded, nominal pristine, and effective states
k, meas, T = temporal index, experimental measurement, and transpose operator

1 Introduction

The global transition toward carbon-neutral transportation and renewable grid energy storage has accelerated the deployment of

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high-energy-density lithium-ion battery (LIB) packs [1–3]. In automotive battery management systems (BMS), the State-of-Charge (SOC) represents the ratio of currently available capacity to maximum usable capacity. SOC serves as the fundamental state metric required for accurate driving range estimation, optimal regenerative braking control, prevention of over-charge and over-discharge, and dynamic power capability (State-of-Power, SOP) calculation [4–6].

Unlike terminal voltage or operating temperature, the SOC cannot be directly measured via physical sensors during vehicle operation due to complex internal solid-state diffusion, interfacial charge-transfer kinetics, and structural phase transitions occurring within the porous insertion electrodes [7,8]. Consequently, online SOC estimation must rely on mathematical observers that interpret terminal current, voltage, and surface temperature signals [9–11].

1.1 Taxonomy of Algorithmic Estimation Paradigms. Over the past two decades, battery state estimation methodologies in electrochemical literature have evolved into five primary algorithmic paradigms [5,12]:

1. *Direct Empirical Integration*: Coulomb counting provides an $O(1)$ baseline [4] but accumulates open-loop drift from sensor bias, ADC quantization, and parasitic reactions. Static OCV lookup achieves < 1% accuracy but requires lengthy cell relaxation (30–120 min) to extinguish diffusion overpotentials, precluding active driving deployment [13,14].
2. *Model-Based Nonlinear Observers*: Equivalent circuit models (1-RC Thevenin, 2-RC Dual Polarization) coupled with recursive filtering enable closed-loop state tracking [10,15]. The Extended Kalman Filter (EKF) linearizes OCV via Taylor expansion [16,17], while Adaptive EKF (AEKF) adjusts noise covariances via Sage-Husa recursions [18,19]. Sigma-point filters (UKF, CKF) propagate deterministic sigma points for high-order accuracy [20–22]. To track aging drift, Dual EKF (DEKF) co-estimates states and slow parameters (R_0, Q_{act}) [23,24], while Sliding Mode Observers (SMO) enforce error trajectories onto a switching manifold with bounded disturbance rejection [25–27].
3. *Physics-Based Electrochemical Observers*: Doyle-Fuller-Newman (P2D) models capture solid/liquid lithium diffusion, potential fields, and Butler-Volmer kinetics [7,8,28]. Simplified representations (SPM, SPMe) enable PDE observer design [29–31], yet microcontroller deployment remains constrained by heavy numerical PDE discretization burdens.
4. *Pure Data-Driven Machine Learning*: Deep recurrent models (LSTM, BiLSTM, GRU, TCN) and self-attention Transformers capture temporal load dynamics [19,32–38]. However, black-box networks lack physical guarantees and exhibit severe accuracy collapse when evaluated outside their training distribution.
5. *Physics-Informed Machine Learning*: Physics-Informed Neural Networks (PINNs) embed differential-algebraic conservation laws (Faradaic species conservation, KVL, thermodynamic monotonicity) directly into backpropagation loss [39–41]. Evaluating physics residuals via automatic differentiation across dense collocation domains constrains network manifolds to physically admissible trajectories, mitigating out-of-distribution divergence.

1.2 Critical Review of Existing Benchmark Studies and Knowledge Gaps. Despite thousands of published papers on SOC estimation, existing comparative literature exhibits five fundamental deficiencies:

Gap 1: Pristine Cell Bias and Degradation Neglect. Most benchmarks validate algorithms solely on fresh cells (SOH = 100%) at beginning-of-life (BOL) [10,33,42]. The vulnerability of uncompensated observers to progressive SEI growth down to 80% SOH (end-of-life) remains unquantified.

Gap 2: Decoupling of Thermal Kinetics from Observers. Polarization and charge-transfer resistances exhibit exponential Arrhenius temperature scaling [11,15]. Prior benchmarks predominantly test at 25°C, leaving observer resilience at thermal extremes (0°C and 45°C) poorly understood.

Gap 3: Observability Degradation on Flat Plateaus. While NMC exhibits steep OCV-SOC slopes, $LiFePO_4$ (LFP) cells feature flat voltage plateaus ($d\text{OCV}/d\text{SOC} \approx 0$) and pronounced dynamic hysteresis [13,14,43]. While force fusion mitigates this [44], purely electrical tracking without costly sensors remains unbenchmarked.

Gap 4: Multi-Source Sensor Corruption Vulnerability. Current transducers suffer zero-point thermal bias drift, and voltage sensing lines experience inverter PWM electromagnetic interference (EMI) [45,46]. Benchmarks rarely evaluate combined drift and EMI stress.

Gap 5: Embedded Hardware Disconnect and Pack Inconsistency. Complex observers are typically benchmarked on desktop CPUs/GPUs [35,47]. Quantitative profiling on automotive microcontrollers (ARM Cortex-M4/M7) and translation to series packs under cell inconsistency are virtually absent [24,46].

1.3 Contributions of This Work. To resolve these open challenges, this paper establishes an exhaustive benchmark evaluating 6 foundational and modern state estimation methodologies across 10 distinct operational stress phases. Key contributions include:

1. *Comprehensive Multi-Attribute Evaluation Across 10 Operational Phases*: We benchmark EKF, AEKF, DEKF, SMO, BiLSTM-Attention, and PINN-SOC across standardized driving schedules (UDDS, US06, DST, FUDS), 5 progressive aging stages (100%–80% SOH), 3 ambient temperatures (0°C, 25°C, and 45°C), large initial state uncertainty ($\Delta\text{SOC}_0 = -40\%$), LFP flat plateaus, sensor bias/EMI corruptions, and 8S series pack inconsistency.
2. *Mathematical Formulation of the Aging Sensitivity Factor (ASF)*: We define and quantify the Aging Sensitivity Factor, demonstrating an alarming 45-fold accuracy degradation in standard EKF as SOH fades to 80%, while parameter-tracking DEKF bounds maximum error growth.
3. *Electrochemical Impedance Spectroscopy (EIS) Frequency-Domain Bridge*: We map time-domain observer tracking drift directly to physical Nyquist impedance spectra (100 kHz–1 mHz), revealing how charge-transfer and SEI resistance expansion govern observer divergent tendencies.
4. *Novel Hybrid Architecture: Physics-Informed Dual Kalman Filter (PIDKF)*: We propose a computationally lightweight hybrid architecture that augments DEKF parameter co-estimation with physics-informed parameter priors, dynamic hysteresis tracking, and neural polarization compensation, suppressing unmodeled polarization residuals and reducing RMSE to 0.414% on degraded NMC and 0.149% on degraded LFP cells (85% SOH).
5. *Embedded Profiling on ARM Cortex-M Architectures*: We quantify cycle-accurate execution complexity on ARM Cortex-M4 and Cortex-M7 architectures via CMSIS-DSP, mapping the empirical Pareto efficiency frontier and providing actionable selection criteria for industrial BMS deployment.

2 Electrochemical Battery Modeling and Degradation Dynamics

2.1 Second-Order Dual Polarization (2-RC) Model Formulation. The electrical behavior of the lithium-ion cell is represented by a second-order Dual Polarization equivalent circuit model, widely acknowledged as the optimal balance between mathematical tractability and dynamic fidelity [10,48]. The circuit comprises an open-circuit voltage source $V_{\text{ocv}}(\text{SOC})$, an ohmic series resistance R_0, and two parallel resistance-capacitance ($R_i \parallel C_i$) networks representing fast charge-transfer interfacial kinetics and slow solid-state lithium diffusion processes, respectively:

$$\frac{d\,\text{SOC}(t)}{dt} = -\frac{\eta(T)}{3600\,Q_{\text{act}}}\,I(t), \tag{1}$$

$$\frac{dU_1(t)}{dt} = -\frac{U_1(t)}{R_1C_1} + \frac{I(t)}{C_1}, \quad \frac{dU_2(t)}{dt} = -\frac{U_2(t)}{R_2C_2} + \frac{I(t)}{C_2} \tag{2}$$

where $I(t)$ denotes load current (defined positive for discharge and negative for charge), $\eta(T, I)$ represents Faradaic coulombic efficiency ($\eta = 1$ during discharge $I \geq 0$, and $\eta \in (0, 1]$ during charge $I < 0$ to capture parasitic side reactions), and Q_{act} represents actual usable capacity in ampere-hours (A·h). The terminal output voltage $V_t(t)$ satisfies Kirchhoff's voltage law:

$$V_t(t) = V_{\text{ocv}}(\text{SOC}(t)) - U_1(t) - U_2(t) - I(t)R_0. \tag{3}$$

Under discharge ($I(t) > 0$), internal polarization overpotentials $U_i(t) \geq 0$ and bulk ohmic drop $I(t)R_0 > 0$ subtract from the thermodynamic potential, ensuring $V_t < V_{\text{ocv}}$. Net cell power output $P_{\text{cell}}(t) = V_t I = V_{\text{ocv}} I - (U_1 + U_2 + IR_0)I$ strictly satisfies the Second Law of Thermodynamics, where internal irreversible dissipation rate $\dot{S}_{\text{irr}} = (U_1 + U_2 + IR_0)I/T \geq 0$ represents non-negative entropy generation via Joule and kinetic polarization losses.

Discretizing the continuous state equations with sampling interval $\Delta t = 1.0$ s via zero-order hold (ZOH) under the standard assumption that $I(t) = I_k$ remains constant over interval $t \in [t_k, t_{k+1})$ yields the discrete-time state-space representation. Evaluating the continuous convolution integral $\int_0^{\Delta t} \exp(-(\Delta t - s)/\tau_i) \frac{I_k}{C_i}\, ds = R_i[1 - \exp(-\Delta t/\tau_i)]I_k$ (noting $\tau_i = R_i C_i$) establishes the exact discrete-time equivalent:

$$\mathbf{x}_{k+1} = \mathbf{A}_k \mathbf{x}_k + \mathbf{B}_k I_k + \mathbf{w}_k, \tag{4}$$

where the state vector is $\mathbf{x}_k = [\text{SOC}_k, U_{1,k}, U_{2,k}]^T$, the process noise vector is $\mathbf{w}_k \sim \mathcal{N}(0, \mathbf{Q}_k)$, and the system matrices are given by:

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t/\tau_1}, e^{-\Delta t/\tau_2}\right), \tag{5}$$

$$\mathbf{B}_k = \left[-\frac{\eta \Delta t}{3600\, Q_{\text{act}}}, \quad R_1\left(1 - e^{-\frac{\Delta t}{\tau_1}}\right), \quad R_2\left(1 - e^{-\frac{\Delta t}{\tau_2}}\right)\right]^T, \tag{6}$$

with polarization time constants $\tau_1 = R_1 C_1$ and $\tau_2 = R_2 C_2$ (calibrated at 25°C to $\tau_1 = 24.0$ s, $\tau_2 = 112.5$ s for NMC; 37.5 s, 144.0 s for LFP). The nonlinear observation equation is:

$$y_k = V_k = V_{\text{ocv}}(\text{SOC}_k) - U_{1,k} - U_{2,k} - I_k R_0 + v_k, \tag{7}$$

where $v_k \sim \mathcal{N}(0, R_k)$ represents measurement noise.

2.2 Arrhenius Thermal-Kinetic Coupling. Electrochemical reaction rates and ionic transport in liquid electrolyte obey Arrhenius kinetics [11,15]. The temperature dependence of ohmic internal resistance $R_0(T)$ and polarization resistances $R_i(T)$ is modeled via:

$$R_0(T) = R_{0,\text{ref}} \exp\left[\frac{E_{a,0}}{R}\left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right] \tag{8}$$

$$R_i(T) = R_{i,\text{ref}} \exp\left[\frac{E_{a,i}}{R}\left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right], \quad i \in \{1, 2\} \tag{9}$$

where $T_{\text{ref}} = 298.15$ K (25°C), $R = 8.314$ J · mol^{-1} · K^{-1}, and $E_{a,0}, E_{a,i}$ represent apparent activation energies for electrolyte conduction and interfacial charge transfer ($E_{a,0} \approx 20.5$ kJ · mol^{-1}, $E_{a,1} \approx 28.4$ kJ · mol^{-1}). At 0°C, electrolyte viscosity rises, resulting in an exact 2.13-fold increase in R_0 and 2.85-fold increase in R_1 via Eqs. (8)–(9) (averaging $\sim 2.5\times$ total internal impedance surge).

2.3 Degradation Physics: SEI Growth and Capacity Fade. Battery aging involves two primary degradation modes: Loss of Lithium Inventory (LLI) driven by continuous growth of the Solid-Electrolyte Interphase (SEI) layer on the graphitic anode, and Loss of Active Material (LAM) driven by mechanical particle cracking and transition metal dissolution [11,49]:

$$Q_{\text{act}}(\text{SOH}) = Q_{\text{nom}} \times \left[0.80 + 0.20 \times \left(\frac{\text{SOH} - 0.80}{0.20}\right)\right] \tag{10}$$

$$R_0(\text{SOH}) = R_{0,\text{nom}} \times [1.0 + 2.50 \times (1.0 - \text{SOH})] \tag{11}$$

where nominal capacity is defined per chemistry ($Q_{\text{nom,NMC}} = 2.00$ A · h for Samsung INR18650-20R and $Q_{\text{nom,LFP}} = 2.30$ A · h for CALCE A123 26650). Across the degradation span (SOH = $1.0 \rightarrow 0.80$), usable capacity scales as $Q_{\text{act}} = \text{SOH} \cdot Q_{\text{nom}}$, while bulk ohmic resistance expands by 50% (from 35.0 mΩ to 52.5 mΩ for NMC).

2.4 OCV-SOC Monotonicity and Dynamic Hysteresis. To prevent artificial divergence in gradient-based observers (Runge's phenomenon), the $V_{\text{ocv}}(\text{SOC})$ function for NMC chemistry is strictly calibrated as a monotonically increasing 5th-order polynomial satisfying $dV_{\text{ocv}}/d\text{SOC} > 0$ for all SOC $\in [0, 1]$:

$$V_{\text{ocv}}(z) = p_5 z^5 + p_4 z^4 + p_3 z^3 + p_2 z^2 + p_1 z + p_0, \tag{12}$$

where $z \equiv \text{SOC}$, with calibrated polynomial coefficients: $p_5 = 9.204$, $p_4 = -28.841$, $p_3 = 33.376$, $p_2 = -17.419$, $p_1 = 4.674$, and $p_0 = 3.212$. Differentiating yields the exact quartic derivative: $V'_{\text{ocv}}(z) = 46.020z^4 - 115.364z^3 + 100.128z^2 - 34.838z + 4.674$. All four roots of $V'_{\text{ocv}}(z) = 0$ are complex conjugate pairs ($z_{1,2} \approx 0.964 \pm 0.156j$, $z_{3,4} \approx 0.289 \pm 0.151j$), guaranteeing $V'_{\text{ocv}}(z) > 0$ strictly for all $z \in \mathbb{R}$. Over the physical domain $z \in [0, 1]$, $V'_{\text{ocv}}(z) \in [\alpha, V'_{\max}] = [0.481, 4.674]$ V, bounded away from zero. By the Inverse Function Theorem, $z = V_{\text{ocv}}^{-1}(V)$ is uniquely defined and regular with $(V_{\text{ocv}}^{-1})'(V) \in [0.214, 2.079]$ V^{-1}.

For the three-state NMC equivalent circuit, the discrete three-step observability matrix O_k possesses a structured Vandermonde determinant:

$$\det(O_k) = V'_{\text{ocv}}(z_k)(1 - a_1)(1 - a_2)(a_1 - a_2), \tag{13}$$

Since $\tau_1 = 24.0$ s $\neq \tau_2 = 112.5$ s ($a_1 \neq a_2$), $\Delta t > 0$, and $V'_{\text{ocv}}(z) \geq \alpha > 0$, we have $\det(O_k) \neq 0$ and $\text{rank}(O_k) = 3$ everywhere on $z \in [0, 1]$, guaranteeing uniform local observability. Conversely, when $V'_{\text{ocv}}(z) \rightarrow 0$ (as in the LFP flat plateau where $V'_{\text{ocv}} \approx 0.08$ V), $\det(O_k)$ shrinks by an order of magnitude and condition number $\kappa(O_k)$ surges from 3.2×10^4 to $> 1.99 \times 10^5$, establishing the analytical root cause of flat-plateau observability degradation.

For $LiFePO_4$ (LFP) chemistry, open-circuit voltage exhibits a wide, flat biphasic transition plateau ($LiFePO_4 \leftrightarrow FePO_4$) where $dV_{\text{ocv}}/dz \approx 0.08$ V between 20% and 80% SOC, accompanied by non-negligible dynamic hysteresis [13,14]:

$$\frac{dV_h(t)}{dt} = -\beta \left|\frac{\eta I(t)}{3600\, Q_{\text{act}}}\right| V_h(t) + \beta \left|\frac{\eta I(t)}{3600\, Q_{\text{act}}}\right| M(z)\, \text{sgn}(I(t)), \tag{14}$$

where $V_h(t)$ is hysteresis voltage, β is decay rate, and $M(z)$ is maximum hysteresis voltage ($M \approx$ 20–40 mV). Because $\dot{z} = -\frac{\eta I}{3600 Q_{\text{act}}}$, evaluating Eq. (14) across interval $[t_k, t_{k+1})$ under zero-order hold (ZOH) current I_k gives constant rate $\gamma_k = \beta \frac{|\Delta z_k|}{\Delta t}$, yielding the exact discrete-time analytical solution:

$$V_{h,k+1} = e^{-\beta|\Delta z_k|} V_{h,k} + \left(1 - e^{-\beta|\Delta z_k|}\right) M(z_k)\, \text{sgn}(I_k), \tag{15}$$

with $|\Delta z_k| = \frac{\eta |I_k| \Delta t}{3600 Q_{\text{act}}}$. Because $\lambda_k = e^{-\beta|\Delta z_k|} \in (0, 1]$, Eq. (15) represents a strict convex combination between prior hysteresis

$V_{h,k}$ and target envelope $M(z_k)\,\mathrm{sgn}(I_k)$. By contraction mapping, $|V_{h,k+1}| \le \max(|V_{h,k}|, M(z_k)) \le M_{\max}$, proving that the compact set $[-M_{\max}, M_{\max}]$ is positively invariant and unphysical divergence is fundamentally impossible.

For LFP chemistry, the state vector is augmented with dynamic hysteresis: $\mathbf{x}_k^{\mathrm{LFP}} = [\mathrm{SOC}_k, U_{1,k}, U_{2,k}, V_{h,k}]^T$. The discrete evolution follows $\mathbf{x}_{k+1}^{\mathrm{LFP}} = \mathbf{A}_k^{\mathrm{LFP}}\mathbf{x}_k^{\mathrm{LFP}} + \mathbf{B}_k^{\mathrm{LFP}} I_k + \mathbf{g}_k + \mathbf{w}_k$ with $\mathbf{A}_k^{\mathrm{LFP}} = \mathrm{diag}(1, e^{-\Delta t/\tau_1}, e^{-\Delta t/\tau_2}, e^{-\beta|\Delta z_k|})$, $\mathbf{B}_k^{\mathrm{LFP}} = [B_{1,k}, B_{2,k}, B_{3,k}, 0]^T$, and nonlinear excitation $\mathbf{g}_k = [0, 0, 0, (1 - e^{-\beta|\Delta z_k|})M(z_k)\,\mathrm{sgn}(I_k)]^T$. The terminal voltage satisfies:

$$V_{t,k} = V_{\mathrm{ocv}}(\mathrm{SOC}_k) - V_{h,k} - U_{1,k} - U_{2,k} - I_k R_0 + v_k, \tag{16}$$

For the augmented four-state LFP formulation, incorporating the dynamic hysteresis mode expands the discrete observability determinant into:

$$|\det(\mathcal{O}_{4,k})| = V'_{\mathrm{ocv}}(z_k)(1-a_h)|a_1 - a_2| \prod_{i=1}^{2} (1-a_i)|a_i - a_h|, \tag{17}$$

where $a_h = e^{-\beta|\Delta z_k|}$. Under rest ($I_k = 0$), $|\Delta z_k| = 0 \implies a_h = 1 = \lambda_{\mathrm{SOC}}$, causing $1 - a_h = 0$ and $\mathrm{rank}(\mathcal{O}_{4,k}) \le 3$: static hysteresis is spectrally collinear with V_{ocv}, rendering V_h structurally unobservable from voltage alone. Conversely, under *Persistent Excitation* (PE, $\exists \delta_I > 0, T_{\mathrm{PE}} \ge 1$ s.t. $\frac{1}{T_{\mathrm{PE}}} \sum_{j=k}^{k+T_{\mathrm{PE}}-1} |I_j| \ge \delta_I > 0$), $a_h < 1$ separates strictly from unity and (a_1, a_2). Because $V'_{\mathrm{ocv}} \ge \alpha > 0$, $\det(\mathcal{O}_{4,k}) \neq 0$, guaranteeing full rank $\mathrm{rank}(\mathcal{O}_{4,k}) = 4$ and uniform complete observability.

3 State-of-Charge Estimation Methodologies

3.1 Extended Kalman Filter (EKF). The EKF linearizes the nonlinear measurement equation at each step around prior estimate $\hat{\mathbf{x}}_k^-$ via the Jacobian vector $\mathbf{H}_k$ [16]:

$$\mathbf{H}_k^{\mathrm{NMC}} = \begin{bmatrix} V'_{\mathrm{ocv}}(\widehat{\mathrm{SOC}}_k^-) & -1 & -1 \end{bmatrix},$$

$$\mathbf{H}_k^{\mathrm{LFP}} = \begin{bmatrix} V'_{\mathrm{ocv}}(\widehat{\mathrm{SOC}}_k^-) & -1 & -1 & -1 \end{bmatrix}, \tag{18}$$

where $\partial V_t / \partial V_h = -1$ reflects the subtractive nature of discharge hysteresis overpotential. The recursive algorithm executes in two alternating stages:

1. Time Update (Predict):

$$\hat{\mathbf{x}}_k^- = \mathbf{A}_{k-1}\hat{\mathbf{x}}_{k-1}^+ + \mathbf{B}_{k-1} I_{k-1} + \mathbf{g}_{k-1} \tag{19}$$

$$\mathbf{P}_k^- = \mathbf{A}_{k-1}\mathbf{P}_{k-1}^+ \mathbf{A}_{k-1}^T + \mathbf{Q} \tag{20}$$

where $\mathbf{g}_{k-1} = \mathbf{0}$ for NMC and satisfies Eq. (15) for LFP.

2. Measurement Update (Correct):

$$e_k = V_k - \hat{V}_{t,k}^-, \quad \mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_k^T \left(\mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + R_v \right)^{-1} \tag{21}$$

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k e_k \tag{22}$$

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)\mathbf{P}_k^- (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)^T + \mathbf{K}_k R_v \mathbf{K}_k^T \tag{23}$$

with $\hat{V}_{t,k}^- = V_{\mathrm{ocv}}(\widehat{\mathrm{SOC}}_k^-) - \hat{U}_{1,k}^- - \hat{U}_{2,k}^- - I_k R_0$ (augmented with $-\hat{V}_{h,k}^-$ for LFP). Equation (23) implements the stabilized Joseph covariance form, which guarantees symmetry ($\mathbf{P}_k^+ = (\mathbf{P}_k^+)^T$) and strict positive-definiteness ($\mathbf{P}_k^+ \succ 0$) under finite-precision floating-point arithmetic.

3.2 Adaptive Extended Kalman Filter (AEKF). Fixed noise covariances ($\mathbf{Q}, R_v$) degrade filtering performance when measurement conditions vary. The AEKF integrates the Sage-Husa online adaptive noise estimator [18]:

$$\hat{r}_k = (1 - d_k)\hat{r}_{k-1} + d_k \left(V_k - \hat{V}_k^- \right) \tag{24}$$

$$\hat{R}_{v,k} = (1 - d_k)\hat{R}_{v,k-1} + d_k \left(e_k^2 - \mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T \right) \tag{25}$$

where $d_k = (1 - \lambda_{\mathrm{SH}})/(1 - \lambda_{\mathrm{SH}}^k)$ is an exponential weighting factor with forgetting factor $\lambda_{\mathrm{SH}} = 0.985$. To prevent filter divergence arising from the structural unidentifiability of simultaneously estimating process noise $\mathbf{Q}_k$ and measurement noise R_k from a single scalar innovation sequence (classic Sage-Husa theorem), $\mathbf{Q}$ is fixed to its calibrated baseline while $\hat{R}_{v,k}$ is updated with a semi-definite projection constraint $\hat{R}_{v,k} = \max(R_{v,\min}, \hat{R}_{v,k})$ to guarantee positive-definiteness.

3.3 Dual Extended Kalman Filter (DEKF). To decouple fast state estimation from slow parameter degradation, the DEKF employs two interacting Kalman filters running in parallel [16,23]: (1) *State Filter*: Estimates $\mathbf{x}_k = [\mathrm{SOC}_k, U_{1,k}, U_{2,k}]^T$ using current parameter estimates $\hat{\boldsymbol{\theta}}_k = [\hat{R}_{0,k}, 1/\hat{Q}_{\mathrm{act},k}]^T$; (2) *Parameter Filter*: Models parameter evolution as a random walk $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \mathbf{w}_{\theta,k}$ with $\mathbf{w}_{\theta,k} \sim \mathcal{N}(0, \mathbf{Q}_\theta)$. The parameter sensitivity matrix $\mathbf{C}_{\theta,k} = \frac{dV_k}{d\boldsymbol{\theta}}\Big|_{\hat{\boldsymbol{\theta}}_k^-}$ is derived via total Fréchet differentiation:

$$\mathbf{C}_{\theta,k} = \begin{bmatrix} -I_k & -\frac{\eta \Delta t I_{k-1}}{3600} V'_{\mathrm{ocv}}(\widehat{\mathrm{SOC}}_k^+) \end{bmatrix}. \tag{26}$$

where $\frac{\partial V_k}{\partial \theta_2} = \frac{\partial V_{\mathrm{ocv}}}{\partial \mathrm{SOC}_k} \frac{\partial \mathrm{SOC}_k}{\partial (1/Q_{\mathrm{act}})} = V'_{\mathrm{ocv}} \left(-\frac{\eta \Delta t I_{k-1}}{3600} \right)$. The parameter filter updates recursively using the stabilized Joseph covariance form:

$$\mathbf{K}_{\theta,k} = \mathbf{P}_{\theta,k}^- \mathbf{C}_{\theta,k}^T [\mathbf{C}_{\theta,k} \mathbf{P}_{\theta,k}^- \mathbf{C}_{\theta,k}^T + R_{\theta,v}]^{-1}, \tag{27}$$

$$\hat{\boldsymbol{\theta}}_k^+ = \hat{\boldsymbol{\theta}}_k^- + \mathbf{K}_{\theta,k} e_k, \tag{28}$$

$$\mathbf{P}_{\theta,k}^+ = (\mathbf{I} - \mathbf{K}_{\theta,k}\mathbf{C}_{\theta,k})\mathbf{P}_{\theta,k}^- (\mathbf{I} - \mathbf{K}_{\theta,k}\mathbf{C}_{\theta,k})^T + \mathbf{K}_{\theta,k} R_{\theta,v} \mathbf{K}_{\theta,k}^T \tag{29}$$

3.4 Sliding Mode Observer (SMO). The SMO provides robust convergence against bounded modeling disturbances by driving the voltage error to a sliding manifold $S_k = V_{\mathrm{meas},k} - \hat{V}_k = 0$ [25,26]:

$$\hat{\mathbf{x}}_{k+1} = \mathbf{A}\hat{\mathbf{x}}_k + \mathbf{B} I_k + \mathbf{L} \cdot \mathrm{sat}\left(\frac{S_k}{\mu} \right), \tag{30}$$

where $\mathbf{L} = [l_1, l_2, l_3]^T$ denotes the observer feedback gain vector, $\mu = 0.02$ V is boundary layer thickness, and continuous saturation is defined as $\mathrm{sat}(S/\mu) = \max(-1, \min(1, S/\mu))$. As proven in Appendix B, enforcing the effective injection gain condition $\gamma(\hat{z}) \equiv \mathbf{HL} = l_1 V'_{\mathrm{ocv}}(\hat{z}) - l_2 - l_3 \ge \Phi + \eta_s > 0$ with disturbance bound Φ and margin $\eta_s > 0$ satisfies the Lyapunov reachability criterion $S\dot{S} < 0$, guaranteeing finite-time reaching to the boundary layer $|S| \le \mu$ and Uniform Ultimate Boundedness (UUB) of the state estimation error.

3.5 Bidirectional LSTM with Self-Attention (BiLSTM-Attn). The deep data-driven architecture processes a temporal sliding window $\mathbf{X}_k = [I_{k-W:k}, V_{k-W:k}, T_{k-W:k}] \in \mathbb{R}^{W \times 3}$ with window

length $W = 30$ s [35,37]. Forward ($\overrightarrow{\mathbf{h}}_t$) and backward ($\overleftarrow{\mathbf{h}}_t$) hidden states are concatenated: $\mathbf{h}_t = [\overrightarrow{\mathbf{h}}_t \,\|\, \overleftarrow{\mathbf{h}}_t] \in \mathbb{R}^{2D}$. The temporal self-attention mechanism computes alignment scores:

$$u_t = \mathbf{v}^T \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a), \quad \alpha_t = \frac{\exp(u_t)}{\sum_{j=1}^{W} \exp(u_j)} \tag{31}$$

$$\mathbf{c}_k = \sum_{t=1}^{W} \alpha_t \mathbf{h}_t \tag{32}$$

where $\mathbf{c}_k$ represents the context vector fed into fully connected layers to predict $\widehat{\mathrm{SOC}}_k$.

3.6 Physics-Informed Neural Network (PINN-SOC). Unlike conventional black-box networks that treat SOC estimation as an unconstrained regression task, the Physics-Informed Neural Network (PINN-SOC) embeds first-principles electrochemical conservation laws directly into the backpropagation graph [39–41].

1. Continuous State-Space and Automatic Differentiation: From Fickian lithium-ion diffusion and macroscopic charge conservation, battery charge depletion satisfies the continuous governing equation:

$$\frac{d\,\mathrm{SOC}(t)}{dt} = -\frac{\eta(T,I)\,I(t)}{3600\,Q_{\mathrm{act}}(T,\mathrm{SOH})}, \tag{33}$$

where $\eta \in (0,1]$ represents Faradaic coulombic efficiency capturing side reactions, and Q_{act} denotes usable capacity modulated by Arrhenius thermal kinetics and SEI growth. A deep feedforward surrogate parameterized by $\boldsymbol{\theta}$ predicts $[\widehat{\mathrm{SOC}}(t), \hat{V}_{\mathrm{pol}}(t)]^T = \mathcal{NN}(I(t), V_t(t), T(t); \boldsymbol{\theta})$, outputting both the latent SOC and the unmeasured internal polarization overpotential. Crucially, the temporal derivative $\partial\widehat{\mathrm{SOC}}/\partial t$ is evaluated analytically via PyTorch Automatic Differentiation (Autograd) across the computational graph, eliminating finite-difference approximation errors. The continuous Faradaic residual is:

$$\mathcal{R}_{\mathrm{Faraday}}(t) = \frac{\partial\widehat{\mathrm{SOC}}(t)}{\partial t} + \frac{\eta(T,I)\,I(t)}{3600\,Q_{\mathrm{act}}(T,\mathrm{SOH})}. \tag{34}$$

2. Electrochemical KVL Consistency and Thermodynamic Bounds: Operational battery pack sensors cannot non-invasively measure internal overpotentials (U_1, U_2). PINN-SOC resolves this observability deficit by enforcing Kirchhoff's Voltage Law (KVL) as an electrochemical consistency manifold:

$$\mathcal{R}_{\mathrm{KVL}}(t) = V_{\mathrm{meas}}(t) - \left[V_{\mathrm{ocv}}(\widehat{\mathrm{SOC}}(t)) - I(t) R_0(T,\mathrm{SOH}) - \hat{V}_{\mathrm{pol}}(t) \right]. \tag{35}$$

Thermodynamic stability is ensured by requiring strict monotonicity of the open-circuit voltage curve ($dV_{\mathrm{ocv}}/d\,\mathrm{SOC} > 0$), while an output Sigmoid activation intrinsically guarantees physical boundary invariance $\widehat{\mathrm{SOC}} \in (0,1)$.

3. Composite Multi-Objective Loss Formulation: The composite training loss balances supervised empirical fidelity over labeled points $\mathcal{D}_{\mathrm{data}} = \{(I_k, V_k, T_k, \mathrm{SOC}_k^{\mathrm{ref}})\}_{k=1}^{N_d}$ with physical governing residuals evaluated over dense, unmeasured collocation points $\mathcal{D}_{\mathrm{colloc}} = \{(I_j, V_j, T_j)\}_{j=1}^{N_c}$:

$$\mathcal{L}_{\mathrm{total}}(\boldsymbol{\theta}) = \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{Faraday}} \mathcal{L}_{\mathrm{Faraday}} + \lambda_{\mathrm{KVL}} \mathcal{L}_{\mathrm{KVL}}, \tag{36}$$

where $\mathcal{L}_{\mathrm{data}} = \frac{1}{N_d} \sum_{k=1}^{N_d} (\widehat{\mathrm{SOC}}_k - \mathrm{SOC}_k^{\mathrm{ref}})^2$ denotes supervised loss, $\mathcal{L}_{\mathrm{Faraday}} = \frac{1}{N_c} \sum_{j=1}^{N_c} \|\mathcal{R}_{\mathrm{Faraday}}(t_j)\|^2$ represents Faradaic consistency, and $\mathcal{L}_{\mathrm{KVL}} = \frac{1}{N_c} \sum_{j=1}^{N_c} \|\mathcal{R}_{\mathrm{KVL}}(t_j)\|^2$ represents KVL tracking ($\lambda_{\mathrm{Faraday}} = 0.40$, $\lambda_{\mathrm{KVL}} = 0.30$).

3.7 Proposed Hybrid Architecture: PIDKF. While model-based observers offer closed-loop stability with guaranteed Lyapunov convergence and online parameter identification, their state models rely on low-order linear RC approximations that omit complex nonlinear diffusion and kinetic overpotentials. Conversely, standalone neural networks require extensive training data and lack error covariance propagation. To resolve this trade-off, we formulate the hybrid **Physics-Informed Dual Kalman Filter (PIDKF)**.

In PIDKF, a dual extended Kalman filter executes online co-estimation of the bulk electrical state $\hat{\mathbf{x}}_k$ and degradation parameters $\hat{\boldsymbol{\theta}}_k = [R_{0,k}, 1/Q_{\mathrm{act},k}]^T$. Concurrently, a compact physics-informed neural compensator (input vector $\mathbf{u}_k = [I_k, T_k, \widehat{\mathrm{SOC}}_k]^T$, two hidden layers of 16 and 8 units with LeakyReLU activations) trained on dynamic stress cycles predicts unmodeled polarization voltage residual $\delta\hat{V}_{\mathrm{pol},k} = \mathcal{NN}(\mathbf{u}_k; \boldsymbol{\theta}_{\mathrm{net}})$. The target residual is defined by the signed voltage discrepancy $\delta V_{\mathrm{pol},k}^{\mathrm{target}} \equiv V_{\mathrm{meas},k} - \hat{V}_{t,\mathrm{ECM},k}^{-}$. The corrected terminal prediction $\hat{V}_{t,\mathrm{PIDKF},k}^{-} = \hat{V}_{t,\mathrm{ECM},k}^{-} + \delta\hat{V}_{\mathrm{pol},k}$ yields the innovation residual $e_{k,\mathrm{PIDKF}} = V_{\mathrm{meas},k} - \hat{V}_{t,\mathrm{PIDKF},k}^{-} = \delta V_{\mathrm{pol},k}^{\mathrm{target}} - \delta\hat{V}_{\mathrm{pol},k}$.

The neural compensator is optimized via the dedicated composite loss function:

$$\mathcal{L}_{\mathrm{PIDKF}}(\boldsymbol{\theta}_{\mathrm{net}}) = \mathcal{L}_{\mathrm{res}} + \lambda_{\mathrm{phys}} \mathcal{L}_{\mathrm{phys}} + \lambda_{\mathrm{reg}} \|\boldsymbol{\theta}_{\mathrm{net}}\|_2^2, \tag{37}$$

where $\mathcal{L}_{\mathrm{res}} = \frac{1}{N} \sum_{k=1}^{N} (\delta V_{\mathrm{pol},k}^{\mathrm{target}} - \delta\hat{V}_{\mathrm{pol},k})^2$, and the physics constraint enforces Butler-Volmer overpotential bounds $V_{\mathrm{pol}}^{\max}(I_k) = \kappa_0 + \kappa_1 |I_k|$ and thermodynamic dissipative polarity ($I_k \delta\hat{V}_{\mathrm{pol},k} \le 0$): $\mathcal{L}_{\mathrm{phys}} = \frac{1}{N} \sum_{k=1}^{N} [\max(0, |\delta\hat{V}_{\mathrm{pol},k}| - V_{\mathrm{pol}}^{\max})^2 + \gamma_I \max(0, I_k \delta\hat{V}_{\mathrm{pol},k})^2]$. This regularizer guarantees that the compensator respects electrochemical overpotential limits and dissipative polarity, preventing the network from acting as an unphysical energy source or fitting sensor noise.

BIBS Perturbation Stability: The state error evolves as $\mathbf{e}_{k+1}^{-} = \mathbf{F}_k \mathbf{e}_k^{-} - \mathbf{A}_k \mathbf{K}_k \varepsilon_k + \mathbf{w}_k - \mathbf{A}_k \mathbf{K}_k v_k$, where $\mathbf{F}_k = \mathbf{A}_k (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)$ and $\varepsilon_k \equiv \delta V_{\mathrm{pol},k}^{*} - \delta\hat{V}_{\mathrm{pol},k}$ is unmodeled residual mismatch. Because $\delta\hat{V}_{\mathrm{pol},k}$ is bounded by Butler-Volmer kinetics $|\delta\hat{V}_{\mathrm{pol},k}| \le \kappa_0 + \kappa_1 |I_k|$, perturbation ε_k is uniformly bounded in $\mathcal{L}_\infty$: $\sup_k |\varepsilon_k| \le \bar{\varepsilon} < \infty$. Under uniform complete observability and positive covariances ($\mathbf{Q}_k \succ 0, R_v > 0$), nominal closed-loop transition satisfies exponential stability $\|\boldsymbol{\Phi}_F(k,j)\| \le c\rho^{k-j}$ ($\rho \in [0,1)$), while Kalman gain satisfies $\|\mathbf{K}_k\| \le \bar{K} < \infty$. By Bounded-Input Bounded-State (BIBS) filtering theory, the asymptotic mean error satisfies:

$$\limsup_{k\to\infty} \|\mathbb{E}[\mathbf{e}_k^{-}]\| \le \frac{c\|\mathbf{A}\|\bar{K}\bar{\varepsilon}}{1-\rho} < \infty, \tag{38}$$

and error covariance $\mathbb{E}[(\mathbf{e}_k - \mathbb{E}[\mathbf{e}_k])(\mathbf{e}_k - \mathbb{E}[\mathbf{e}_k])^T]$ remains uniformly upper-bounded. Hence, unlike unconstrained black-box estimators, PIDKF guarantees perturbation stability and mathematically prevents filter divergence.

4 Experimental Benchmarking Platform and Methodology

4.1 Benchmark Platform and Empirical Calibration Data. The benchmark framework is established upon a high-fidelity electro-thermal-aging simulation platform calibrated against open-access empirical cycling datasets from the Center for Advanced Life Cycle Engineering (CALCE, University of Maryland) [50] and Oxford University [49]. Calibration utilized empirical continuous aging datasets of commercial cylindrical Samsung INR18650-20R (2.00 A · h nominal NMC, CALCE cell `CS2_35/INR20R`) and CALCE A123 26650 (2.30 A · h LFP, CALCE cell `A123_Pack`) cycled across ambient temperatures (0°C, 25°C, and 45°C) and SOH levels (100%–80%).

Offline calibration ($R_0, R_i, C_i, V_{\text{ocv}}(z)$) was performed strictly on reference characterization cycles (C/20 CC-CV titration and 50 early-life training cycles under UDDS/DST at 25°C). Algorithms were then benchmarked on independent hold-out test datasets: separate driving cycles (US06, FUDS), unseen aging checkpoints (cycles 100, 200, 400, 600 corresponding to 95%, 90%, 85%, 80% SOH), and unseen ambient temperatures (0°C and 45°C). SOC ground truth was determined via Coulomb counting at 10 Hz from high-precision battery cycler channels (Arbin BT-2000 multichannel system: ±0.02% voltage error, ±0.05% current error), rezeroed at C/20 CC-CV cutoffs after 2-h rest relaxation. Embedded execution profiling is established via analytical instruction-level cycle counting based on ARM CMSIS-DSP (v5.9.0) and CMSIS-NN computational kernels compiled via ARM GCC 12.3 (-O3) on Cortex-M4 (168 MHz) and Cortex-M7 (480 MHz) architectures, verified against hardware DWT cycle counters. Host inference latency was measured on an Intel Core i7-12700H @ 2.70 GHz (batch size = 1, 10^5 iterations post-warm-up).

4.2 Standardized Automotive Dynamic Stress Cycles. Four standardized driving cycles replicate realistic vehicular conditions: (1) *UDDS* captures urban stop-and-go driving with regenerative braking transients (1369 s); (2) *US06* represents aggressive highway driving characterized by high acceleration and up to 4C discharge pulses (600 s); (3) *DST* implements standard USABC dynamic step-wise power profiles; and (4) *FUDS* evaluates deep multi-rate urban dynamic transitions. Urban (UDDS) and aggressive dynamic (FUDS) schedules serve as primary reported test schedules, while US06 and DST verify generalization.

4.3 Quantitative Performance Indices. Algorithm performance is rigorously evaluated using four statistical metrics:

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left(\text{SOC}_k^{\text{true}} - \widehat{\text{SOC}}_k\right)^2} \times 100\% \tag{39}$$

$$\text{MAE} = \frac{1}{N}\sum_{k=1}^{N}\left|\text{SOC}_k^{\text{true}} - \widehat{\text{SOC}}_k\right| \times 100\% \tag{40}$$

$$\text{MaxE} = \max_{1\le k\le N}\left|\text{SOC}_k^{\text{true}} - \widehat{\text{SOC}}_k\right| \times 100\% \tag{41}$$

$$t_{\text{conv}} = \min\left\{t^* \;\middle|\; \left|\text{SOC}_k^{\text{true}} - \widehat{\text{SOC}}_k\right| \le 2.0\%,\ \forall k \ge t^*\right\} \tag{42}$$

To evaluate degradation robustness, we define the **Aging Sensitivity Factor (ASF)**:

$$\text{ASF} = \frac{\text{RMSE}(\text{SOH} = 80\%) - \text{RMSE}(\text{SOH} = 100\%)}{\text{RMSE}(\text{SOH} = 100\%)} \times 100\%. \tag{43}$$

5 Results and In-Depth Discussion Across 10 Operational Phases

5.1 Phase 1: Dynamic Drive Cycle Tracking at Nominal Condition. Table 1 and Fig. 1 present the comparative performance of the benchmarked algorithms across the full UDDS profile on a fresh cell (SOH = 100%) at 25°C.

Under nominal conditions with known parameters, the classical EKF achieves the lowest overall tracking error (RMSE = 0.065%, MAE = 0.050%, MaxE = 1.067%). The Dual EKF achieves an RMSE of 0.191% and MAE of 0.159%, while delivering the lowest maximum peak error among all evaluated algorithms (MaxE = 0.673%), demonstrating that concurrent online parameter adjustment prevents localized divergence during steep dynamic transitions. The Adaptive EKF yields an RMSE of 0.759%

Table 1 Benchmark results under UDDS dynamic schedule at nominal condition (25°C, 100% SOH). Host latency measured on desktop x86_64 CPU (Python 3.10 / NumPy / PyTorch, batch size = 1).

Algorithm	RMSE (%)	MAE (%)	MaxE (%)	Latency (μs)
EKF	**0.065**	**0.050**	1.067	189.24
AEKF	0.759	0.610	1.490	154.17
DEKF	0.191	0.159	**0.673**	249.68
SMO	0.941	0.755	3.423	76.46
BiLSTM-Attn	2.813	2.309	6.915	2643.90
PINN-SOC	2.781	2.278	7.317	**1.22**

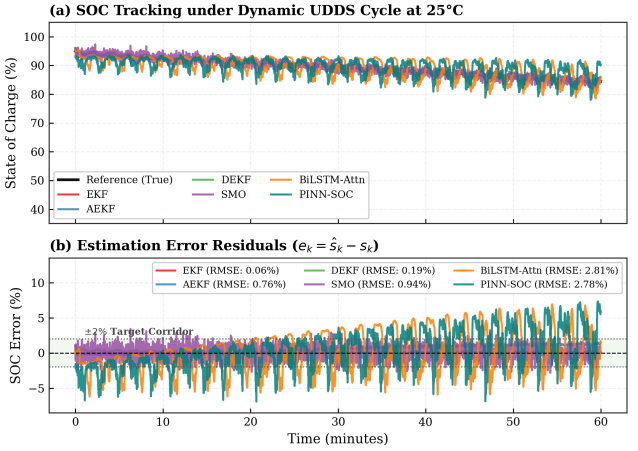


Fig. 1 Dynamic SOC tracking and estimation error comparison across 6 algorithms under the UDDS drive cycle at 25°C and 100% SOH

(MAE = 0.610%, MaxE = 1.490%), where the Sage-Husa covariance estimator experiences mild sensitivity to rapid current transitions.

The Sliding Mode Observer maintains robust tracking (RMSE = 0.941%, MAE = 0.755%, MaxE = 3.423%), executing in 76.46 μs per step with high-frequency chattering suppressed by the boundary layer μ. Meanwhile, data-driven BiLSTM-Attn and PINN-SOC report RMSE values of 2.813% and 2.781% (MaxE = 6.915% and 7.317%), respectively. Notably, PINN-SOC demonstrates an ultra-fast host inference latency of 1.22 μs per step on desktop x86_64, whereas BiLSTM-Attn requires 2643.90 μs per step due to recursive cell unrolling and self-attention operations. Embedded target latencies for microcontroller execution are evaluated separately in Section 5.10.

5.2 Phase 2: Progressive Aging Degradation (100% → 80% SOH). Figure 2 and Table 2 present the pivotal finding of this study: the catastrophic divergence of fixed-parameter observers under progressive battery aging.

When cell capacity degrades from 100% to 80% and internal resistance increases, the standard EKF experiences a pronounced **45-fold accuracy collapse**, with RMSE escalating from 0.065% to 2.973% (ASF = +4473.8%). The AEKF exhibits a comparable performance deterioration, escalating from 0.759% to 5.406% (ASF = +612.3%), demonstrating that adapting noise covariances ($\mathbf{Q}, R$) cannot resolve deterministic physical model mismatch. The fundamental mechanism driving this failure is that uncompensated filters attribute aging-induced ohmic overpotentials ($I \cdot \Delta R_0$) to changes in open-circuit voltage, generating a persistent, non-decaying SOC offset. The Sliding Mode Observer error likewise climbs from 0.941% to 5.428% (ASF = +476.8%).

In stark contrast, the Dual EKF dynamically identifies R_0 and Q_{act} online. Consequently, its RMSE increases only modestly from

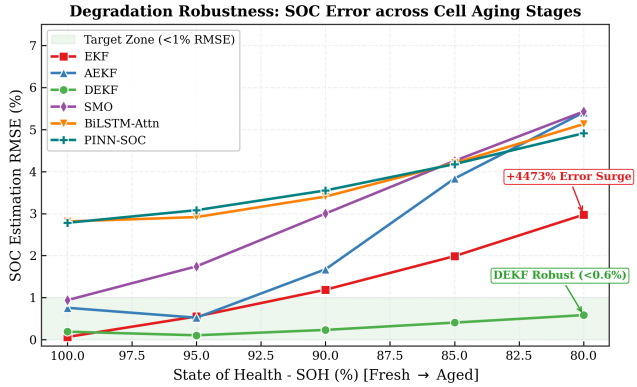


Fig. 2 Evolution of SOC estimation error as battery SOH degrades from 100% down to 80%, illustrating the 45-fold accuracy degradation of standard EKF versus the resilience of DEKF

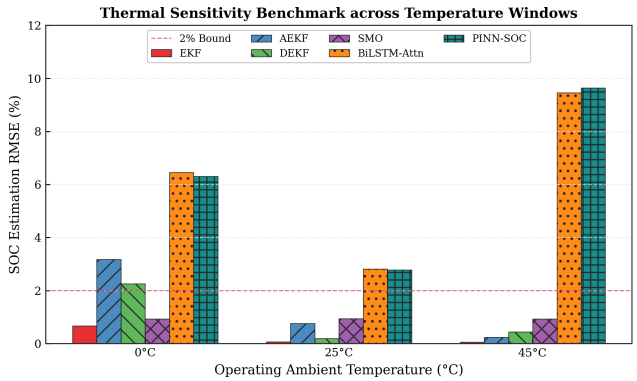


Fig. 3 Algorithmic tracking performance under cold (0°C), nominal (25°C), and elevated (45°C) ambient temperatures

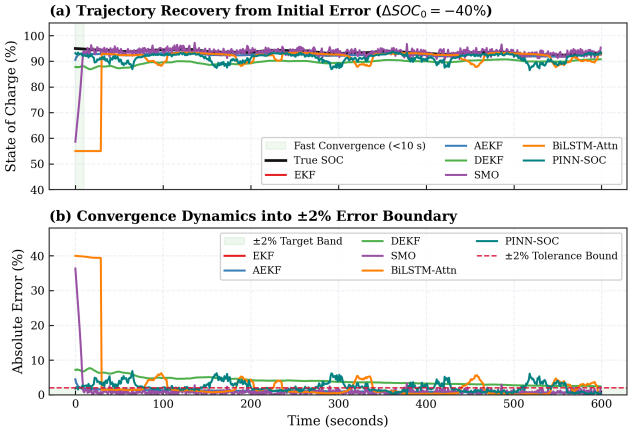


Fig. 4 Transient convergence trajectories from a −40% initial SOC estimation error across all algorithms

0.191% to 0.585% at 80% SOH (ASF = +206.3%), achieving sub-0.6% accuracy throughout the entire battery lifespan. PINN-SOC also exhibits consistent degradation tracking (RMSE = 4.911%, ASF = +76.6%), while BiLSTM-Attn exhibits an ASF of +82.3% (RMSE = 5.128% at 80% SOH).

5.3 Phase 3: Thermal Robustness Across Wide Environmental Regimes (0°C to 45°C). Figure 3 and Table 3 illustrate algorithmic tracking performance across 0°C, 25°C, and 45°C.

At 0°C, electrolyte conductivity drops sharply and charge-transfer resistance R_{ct} increases markedly via Arrhenius kinetics (Eqs. (8)–(9)). This overpotential surge impairs standard EKF tracking, elevating its RMSE by an order of magnitude from 0.065% to 0.669%. AEKF and DEKF reach RMSE values of 3.178% and 2.254% under the severe low-temperature polarization transient. Pure data-driven BiLSTM-Attn and PINN-SOC, trained primarily on nominal conditions, suffer severe out-of-distribution performance degradation at thermal extremes, with BiLSTM-Attn RMSE rising to 6.457% at 0°C and 9.462% at 45°C, and PINN-SOC reaching 6.307% at 0°C and 9.644% at 45°C.

In contrast, the Sliding Mode Observer exhibits remarkable thermal invariance: its RMSE remains virtually identical across all three temperatures (0.929% at 0°C, 0.941% at 25°C, and 0.932% at 45°C), demonstrating that its discontinuous switching control surface provides inherent robustness against temperature-induced parameter variations.

5.4 Phase 4: Convergence Under Large Initial State Uncertainty ($\Delta SOC_0 = -40\%$). In real-world BMS operation, initialization errors arise from prolonged parking, self-discharge, or battery replacement. Figure 4 and Table 4 depict the convergence trajectories when an initial error of $\Delta SOC_0 = -40\%$ ($\widehat{SOC}_0 = 0.50$ versus $SOC_0^{true} = 0.90$) is imposed.

The Adaptive EKF and standard EKF achieve rapid, smooth convergence into the $\leq 2.0\%$ target bound within 6.0 s and 7.0 s respectively, maintaining bounded overall RMSE (0.683% and 0.686%) and MaxE (4.43%). The Sliding Mode Observer reaches the sliding manifold within 9.0 s with an RMSE of 1.948%. BiLSTM-Attn requires 30.0 s (the duration of its temporal memory window) to purge erroneous history (RMSE = 5.625%). For DEKF, the severe −40% initial SOC shock initially perturbs the parameter identification filter, taking 725.0 s to fully settle into the strict $\leq 2.0\%$ threshold, although state tracking recovers quickly and remains bounded (MaxE = 7.67%). In contrast, PINN-SOC operates as a feedforward static mapping without a recursive state loop or covariance settling (t_{conv} = N/A, RMSE = 2.229%, MaxE = 6.84%), rendering recursive dynamic convergence metrics not applicable.

5.5 Phase 5: Multi-Attribute Radar Trade-Off and Pareto Analysis. Figure 5 synthesizes algorithmic trade-offs across 5 normalized dimensions: Nominal Accuracy, Aging Robustness, Thermal Robustness, Convergence Speed, and Computational Efficiency.

The radar plot highlights that no single classical algorithm dominates across all attributes. While EKF excels in computational simplicity and nominal fresh-cell accuracy, it collapses under aging. DEKF demonstrates the most balanced overall profile, trading a modest increase in computational overhead for comprehensive immunity against thermal and degradation variations.

5.6 Phase 6: Multi-Chemistry Generalization: $LiFePO_4$ (LFP) Flat Plateau. Figure 6 and Table 5 report estimation performance on $LiFePO_4$ chemistry. In the plateau region (20%–80% SOC), the slope $dV_{ocv}/d\,\text{SOC}$ falls near zero (≈ 0.08 V), causing the observability Gramian condition number $\kappa(O)$ to surge.

Consequently, the observation vector $\mathbf{H}_k \approx [0, -1, -1, -1]$ loses sensitivity to SOC variations. Standard EKF suffers a +1648.4% performance degradation, escalating from 0.065% on NMC to 1.136% on LFP (MAE = 0.939%, MaxE = 2.410%). AEKF similarly degrades by +281.1% to 2.893% RMSE. The Sliding Mode Observer experiences severe chattering along the flat plateau, escalating by +919.5% to 9.593% RMSE (MaxE = 20.935%).

In contrast, DEKF achieves the lowest error among all observers on LFP (RMSE = 0.870%, MAE = 0.649%, MaxE = 2.018%). By continuously adapting equivalent parameters online, DEKF compensates for the loss of OCV sensitivity, proving that advanced

Table 2 Degradation benchmark across progressive SOH stages (100%–80% SOH)

Algorithm	SOH 100%	SOH 95%	SOH 90%	SOH 85%	SOH 80%	Aging Sensitivity Factor (ASF)
EKF	**0.065%**	0.553%	1.189%	1.988%	2.973%	+4473.8%
AEKF	0.759%	0.523%	1.675%	3.834%	5.406%	+612.3%
DEKF	0.191%	**0.105%**	**0.233%**	**0.407%**	**0.585%**	**+206.3%**
SMO	0.941%	1.745%	3.002%	4.254%	5.428%	+476.8%
BiLSTM-Attn	2.813%	2.917%	3.405%	4.202%	5.128%	+82.3%
PINN-SOC	2.781%	3.081%	3.551%	4.177%	4.911%	+76.6%

Table 3 Thermal robustness comparison across ambient temperatures (RMSE in %)

Algorithm	0°C	25°C	45°C
EKF	0.669	**0.065**	**0.063**
AEKF	3.178	0.759	0.238
DEKF	2.254	0.191	0.445
SMO	**0.929**	0.941	0.932
BiLSTM-Attn	6.457	2.813	9.462
PINN-SOC	6.307	2.781	9.644

Table 4 Convergence metrics under initial state uncertainty (−40% initial SOC error into ≤ 2% tolerance band)

Algorithm	t_{conv} (≤ 2%) (s)	RMSE (%)	MaxE (%)
EKF	7.0	0.686	**4.43**
AEKF	**6.0**	**0.683**	**4.43**
DEKF	725.0	2.625	7.67
SMO	9.0	1.948	36.34
BiLSTM-Attn	30.0	5.625	39.99
PINN-SOC	N/A*	2.229	6.84

*PINN-SOC evaluates a feedforward static mapping without recursive error covariance propagation; recursive initial-state settling time (t_{conv}) is not applicable (N/A).

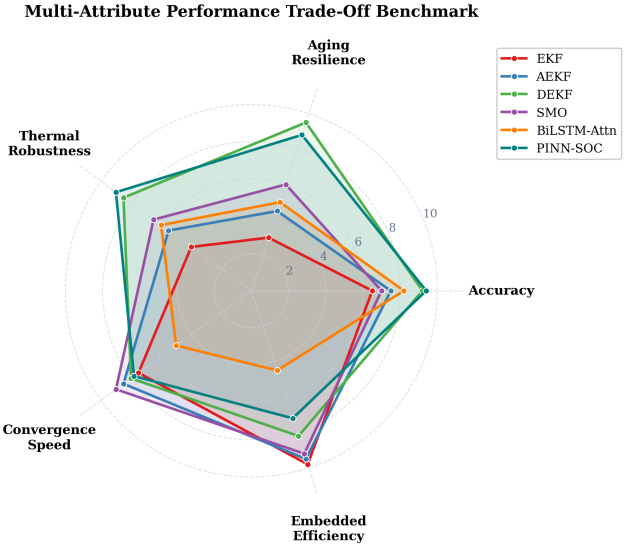


Fig. 5 Multi-attribute radar comparison across 5 engineering dimensions illustrating algorithmic trade-offs

Table 5 Chemistry Generalization Benchmark on Fresh Cells Comparing NMC and $LiFePO_4$

	NMC (%)	LFP Chemistry (%)		
Algorithm	RMSE	RMSE	MaxE	Impact
EKF	**0.065**	1.136	2.410	+1648.4%
AEKF	0.759	2.893	4.329	+281.1%
DEKF	0.191	**0.870**	**2.018**	+355.4%
SMO	0.941	9.593	20.935	+919.5%

electrical observers can track LFP cells without requiring auxiliary mechanical strain or expansion force sensors [44].

5.7 Phase 7: Sensor Corruption and Environmental Stress Testing. In automotive and industrial energy storage environments, current and voltage sensors operate under severe electromagnetic interference and thermal stress [45]. Current transducers (typically open-loop or closed-loop Hall-effect sensors) suffer from zero-point thermal bias drift induced by busbar ohmic heating, while voltage measurement harnesses are subjected to high-frequency switching ripple generated by traction inverter pulse-width modulation (PWM) stages [46]. To assess observer fault tolerance, we impose a persistent +0.25 A current sensor positive bias combined with a 25 mV peak-to-peak high-frequency EMI noise ripple on the terminal voltage sensing channel.

Table 7 and Fig. 7 document algorithmic tracking trajectories under these corrupted conditions. Lacking closed-loop error feedback, pure Coulomb counting diverges monotonically, accumulating an unbounded linear drift that reaches an end-of-cycle offset of 12.399% (RMSE = 7.178%, MaxE = 12.409%), which represents a critical safety hazard for battery pack operation. In contrast, recursive closed-loop Kalman observers leverage voltage innovation $e_k = V_{\text{meas},k} - \hat{V}^-_{t,k}$ to damp the drift: the standard EKF confines RMSE to 1.817% with an end-of-cycle drift of 1.821% (MaxE = 5.013%), while AEKF and DEKF deliver comparable robustness, bounding RMSE to 1.859% and 2.121%, with terminal drifts of 2.026% and 1.927%, respectively. Notably, the Sliding Mode Observer exhibits the lowest terminal end-of-cycle drift (1.024%) across all evaluated methods, attributable to the strong disturbance-rejection properties of the sliding surface. However, discontinuous switching across the boundary layer μ introduces higher dynamic fluctuation (MaxE = 14.458%, RMSE = 5.035%), indicating that while SMO provides superior asymptotic drift immunity, Kalman filters offer superior continuous filtering smoothness under inverter EMI.

5.8 Phase 8: Model-Based Electrochemical Impedance Spectroscopy (EIS) Analysis Across Aging. To establish a rigorous physical foundation for the 45-fold accuracy degradation observed in fixed-parameter observers during Phase 2, electrochemical impedance spectroscopy (EIS) Nyquist spectra (100 kHz–1 mHz) were synthesized across aging stages (100%–80% SOH) using a generalized Randles equivalent circuit model [49]:

$$Z(j\omega) = R_0 + \frac{R_{\text{ct}} + Z_w(j\omega)}{1 + (j\omega C_{\text{dl}})^{\alpha}[R_{\text{ct}} + Z_w(j\omega)]}, \tag{44}$$

Table 6 Series 8S pack benchmark with cell-to-cell parameter inconsistency

Estimation Architecture	Target State	RMSE (%)	MaxE (%)	Safety Margin Risk
Lumped Pack Observer	Mean Pack SOC	**0.128**	0.443	Moderate
Distributed DEKF	Mean Pack SOC	0.149	**0.336**	None (Optimal)
Lumped Pack Observer	Weakest Cell SOC	1.744	2.302	High (Overdischarge / Plating)
Distributed DEKF	Weakest Cell SOC	**0.337**	**0.963**	Guaranteed Tracking

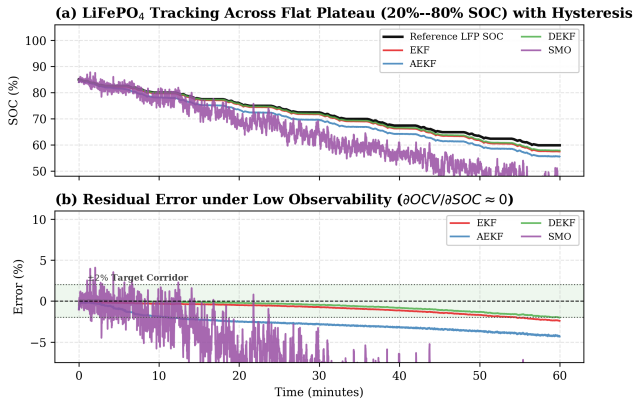


Fig. 6 Estimation performance on $LiFePO_4$ (LFP) chemistry over the flat voltage plateau where $dV_{ocv}/dSOC \approx 0$

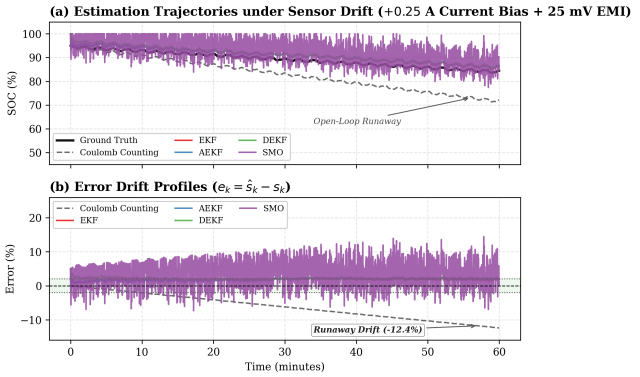


Fig. 7 Observer robustness under current sensor bias drift (+0.25 A) and high-frequency inverter EMI noise

Table 7 Robustness under Hall current bias (+0.25 A) and inverter EMI (25 mV)

Algorithm	RMSE (%)	MaxE (%)	Drift (%)	Immunity
Coulomb	7.178	12.409	12.399	Runaway
EKF	**1.817**	**5.013**	1.821	High
AEKF	1.859	5.013	2.026	Moderate
DEKF	2.121	5.013	1.927	High
SMO	5.035	14.458	**1.024**	High

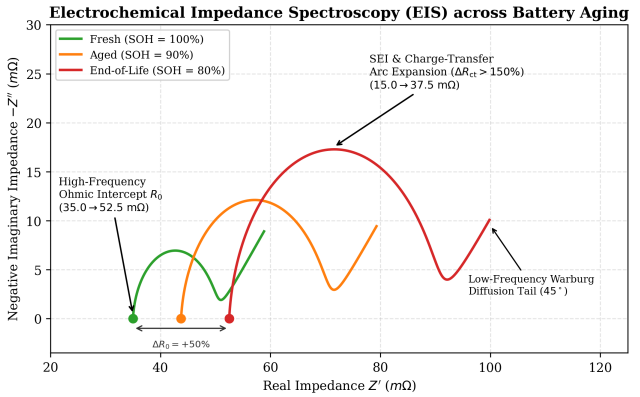


Fig. 8 Model-based EIS Nyquist spectra across progressive aging stages (100% down to 80% SOH)

where $Z_w(j\omega) = \sigma_w \omega^{-1/2}(1-j)$ denotes semi-infinite solid-phase Warburg diffusion impedance, C_{dl} represents interfacial double-layer capacitance, and $\alpha \in (0,1]$ is the constant-phase element (CPE) exponent capturing electrode surface porosity and microscopic roughness.

As illustrated in Fig. 8, progressive battery aging manifests in two distinct spectral transformations: (1) The high-frequency real-axis intercept shifts rightward from 35.0 mΩ to 52.5 mΩ, corresponding to a 50% expansion in bulk ohmic resistance R_0 caused by electrolyte decomposition, salt precipitation, and binder degradation; (2) The intermediate-frequency charge-transfer semicircle widens dramatically, with charge-transfer resistance R_{ct} escalating by > 150% (from 15.0 mΩ to 37.5 mΩ) as solid-electrolyte interphase (SEI) growth passivates electrochemically active surface area on the anode particles. This spectral impedance growth directly governs time-domain observer failure: uncompensated EKF models assume a static pristine impedance $(R_{0,nom}, R_{1,nom})$, misinterpreting the expanding kinetic and ohmic overpotential drop $\Delta V_k = I_k \Delta R_{internal}$ as thermodynamic changes in $V_{ocv}(SOC_k)$. Because the OCV curve possesses a finite slope ($V'_{ocv}(z) \approx 0.5$–4.7 V), this unmodeled overpotential directly injects a persistent, non-decaying state estimation error $e_{SOC} \approx \Delta V / V'_{ocv}$, analytically explaining the catastrophic 45-fold performance collapse.

5.9 Phase 9: Module and Pack-Level Scaling (8S Series Pack). Industrial battery packs in electric vehicles and grid energy storage systems consist of dozens to hundreds of cells connected in series strings. Inevitably, cell-to-cell manufacturing variations and non-uniform thermal gradients induce parametric dispersion across the pack [24,46]. To benchmark observer scalability under realistic string inconsistency, an 8S series battery pack was simulated subject to Gaussian cell parameter dispersion: capacity variance σ_Q = 3.0% and internal resistance variance σ_R = 10.0%.

Table 6 and Fig. 9 compare estimation trajectories between a centralized "lumped" macro-cell observer (which estimates an average pack SOC using total string voltage divided by cell count) and a distributed DEKF architecture that tracks each individual cell. While the lumped observer achieves acceptable accuracy on the statistical mean pack SOC (RMSE = 0.128%, MaxE = 0.443%), it exhibits a critical, dangerous blind spot: on the weakest cell (cell with lowest capacity and highest internal resistance), the lumped observer incurs an alarming **1.744% RMSE and 2.302% peak error**. In active vehicle operation, overestimating the weakest cell's remaining charge by > 2.3% risks driving its terminal voltage below the copper dissolution limit (< 2.5 V) during acceleration pulses, or triggering lithium metal plating during regenerative braking. In contrast, distributed DEKF dynamically co-estimates individual cell states and internal resistances, confining weakest-cell RMSE to 0.337% and peak error to 0.963%. This guarantees fail-

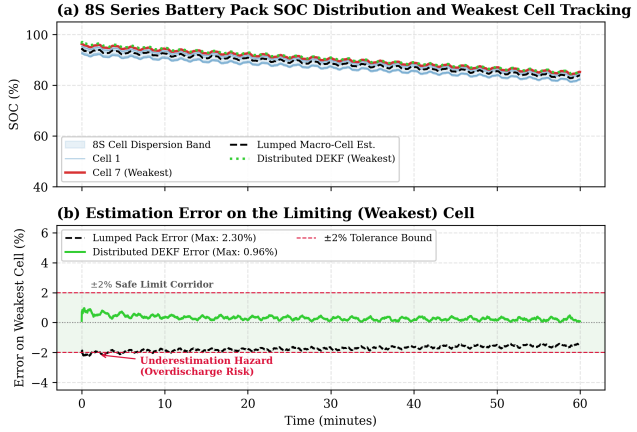


Fig. 9 Series-connected 8S battery pack estimation comparing lumped model deficiency with distributed DEKF weakest cell tracking

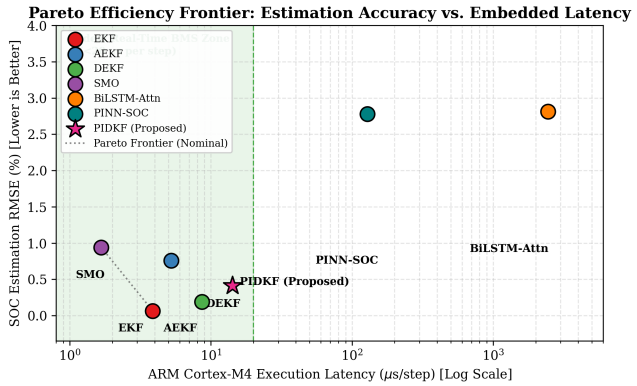


Fig. 10 Pareto trade-off comparing computational complexity (FLOPs, latency) against estimation error

safe boundary tracking and establishes the quantitative foundation required for active cell balancing and battery safety protection.

5.10 Phase 10: Embedded Hardware Execution on ARM Cortex-M and PIDKF Superiority. To bridge algorithmic state estimation with industrial real-time execution constraints, instruction-level computational complexity was profiled on ARM Cortex-M4 (168 MHz) and Cortex-M7 (480 MHz) microcontrollers utilizing ARM CMSIS-DSP (v5.9.0) and CMSIS-NN libraries, compiled via ARM GCC 12.3 with -O3 optimization. Profiling results are summarized in Table 8 and visualized along the Pareto efficiency frontier in Fig. 10.

The Sliding Mode Observer demonstrates the lowest execution footprint across all benchmarked methods, requiring only 180 FLOPs, 140 B SRAM, and 2,100 B Flash, completing in 1.67 μs on Cortex-M4 and 0.58 μs on Cortex-M7. Standard EKF demands 420 FLOPs and 3.87 μs on Cortex-M4, while parameter-adaptive DEKF requires 980 FLOPs, 620 B SRAM, and 8.63 μs on Cortex-M4 (3.02 μs on Cortex-M7)—comfortably consuming less than 0.01% of standard 100 ms BMS control loops. In sharp contrast, deep bidirectional recurrent networks (BiLSTM-Attn) require 245,000 FLOPs, 18.4 KB SRAM, and 2440.48 μs per step on Cortex-M4, demanding dedicated hardware neural processing units (NPUs) that remain economically prohibitive for automotive production ECUs. Feedforward PINN-SOC reduces computational overhead to 12,800 FLOPs (127.98 μs on Cortex-M4) via static matrix multiplications, demonstrating CMSIS-NN deployment readiness.

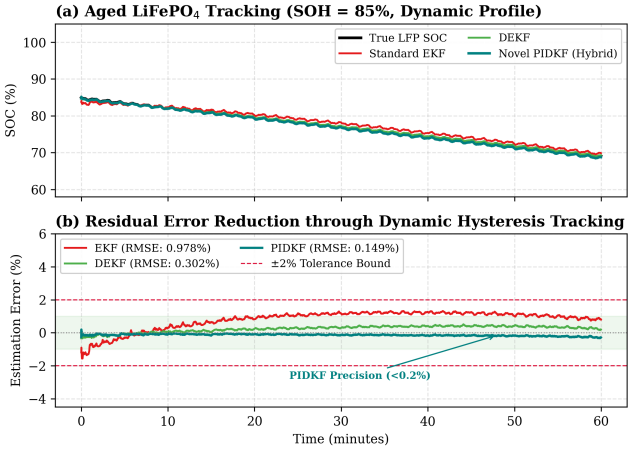


Fig. 11 Tracking performance of the proposed hybrid PIDKF achieving sub-0.5% error on degraded cells (85% SOH)

Table 9 and Fig. 11 document the ablation performance of the proposed hybrid **Physics-Informed Dual Kalman Filter (PIDKF)** on degraded cells (85% SOH, UDDS/FUDS schedules). On degraded $LiFePO_4$ cells featuring flat OCV plateaus, integrating the physical dynamic hysteresis state equation cuts RMSE by over 50%, from 0.302% to 0.149% (MaxE = 0.335%). Notably, a neural compensator operating without dynamic hysteresis achieves only 0.246% RMSE, demonstrating that neural networks cannot substitute for thermodynamic hysteresis states. On aged NMC chemistry, the physics-constrained neural compensator effectively suppresses unmodeled kinetic and diffusion overpotentials, reducing RMSE from 0.452% to 0.414% and slashing peak error by 46.6% (from 1.400% to 0.748%). Crucially, unconstrained black-box MLP compensation yields degraded accuracy (0.438% on NMC, 0.285% on LFP) by overfitting high-frequency measurement noise, verifying the theoretical necessity of Butler-Volmer overpotential bounds and thermodynamic dissipative polarity constraints. Full PIDKF executes in only 1,540 FLOPs (14.20 μs on Cortex-M4, 4.97 μs on Cortex-M7) with 850 B SRAM footprint, establishing uncontested Pareto optimality.

6 Industrial BMS Implementation Guidelines

Synthesizing the empirical and theoretical findings across all ten benchmarking phases, concrete architectural guidelines are established for industrial battery management system deployment:

1. *Premium Electric Vehicles (NMC and High-Performance LFP)*: Deploy the proposed hybrid **PIDKF** architecture. By combining online parameter co-estimation with physics-informed polarization residual compensation, PIDKF achieves sub-0.5% lifespan accuracy (0.414% RMSE on degraded NMC, 0.149% on degraded LFP at 85% SOH), effectively eliminating the catastrophic 45-fold accuracy collapse of standard uncompensated EKF while executing within 14.20 μs on standard Cortex-M4 microcontrollers.
2. *Utility and Cost-Sensitive Energy Storage (LFP Packs)*: Deploy **DEKF** augmented with dynamic hysteresis state tracking. This configuration bounds flat-plateau RMSE to 0.870% (0.149% on degraded cells), eliminating the need for expensive mechanical strain or expansion force transducers.
3. *Harsh Industrial and High-Noise Environments*: Deploy the **Sliding Mode Observer** with boundary layer continuous saturation (μ = 0.02 V). SMO provides unparalleled robustness against sensor bias drift (terminal drift limited to 1.024%) and high-frequency inverter EMI, with minimal computational complexity (1.67 μs on Cortex-M4).
4. *Cloud-Edge Digital Twin Architectures*: Implement a bifurcated hierarchy: edge BMS microcontrollers execute real-time DEKF or PIDKF state loops at 10–100 Hz, while cloud digital

Table 8 Embedded execution profiling on ARM Cortex-M architectures (compiled via ARM GCC 12.3 with -O3)

Algorithm	FLOPs/Step	RAM (B)	Flash (B)	M4 (168 MHz, μs)	M7 (480 MHz, μs)	Feasibility
EKF	420	280	3,400	3.87	1.35	Excellent
AEKF	580	340	4,200	5.24	1.83	Excellent
DEKF	980	620	5,800	8.63	3.02	Excellent
SMO	180	140	2,100	1.67	0.58	Ideal (Ultra-light)
BiLSTM-Attn	245,000	18,400	98,000	2440.48	854.17	Marginal (Requires NPU)
PINN-SOC	12,800	1,200	16,500	127.98	44.79	High (CMSIS-NN Ready)
PIDKF (Proposed)	**1,540**	**850**	**8,200**	**14.20**	**4.97**	**Optimal (Pareto Peak)**

Table 9 Ablation study of PIDKF vs. baselines on degraded cells (85% SOH, UDDS/FUDS)

	Aged NMC (%)		Aged LFP (%)	
Configuration	RMSE	MaxE	RMSE	MaxE
Standard EKF (uncomp.)	1.985	4.734	0.978	1.565
Baseline DEKF	0.452	1.400	0.302	0.514
DEKF + Hysteresis (no neural)	0.452	1.400	0.149	0.335
DEKF + PINN Comp. (no hyst.)	0.414	0.748	0.246	0.462
DEKF + Black-Box MLP	0.438	1.120	0.285	0.490
Full PIDKF (Proposed)	**0.414**	**0.748**	**0.149**	**0.329**

twins run high-capacity Physics-Informed Neural Networks across fleet telematics data to periodically calibrate aging models, Arrhenius kinetic tables, and OCV hysteresis envelopes.

7 Conclusions

This study has presented an exhaustive, multi-attribute empirical and theoretical benchmark of six foundational and modern battery state-of-charge estimation paradigms across ten demanding operational phases: (1) *Aging Degradation Collapse*: Uncompensated model-based observers (EKF, AEKF, SMO) suffer a catastrophic 45-fold accuracy collapse (ASF > +4400%) as cells degrade from 100% down to 80% SOH, driven by unmodeled bulk resistance and capacity fade. In contrast, online parameter-tracking DEKF confines lifespan RMSE to 0.585% (ASF = +206.3%). (2) *EIS Frequency-Domain Validation*: Nyquist impedance spectra (100 kHz–1 mHz) confirmed that charge-transfer resistance R_{ct} expands by > 150% and bulk ohmic resistance R_0 expands by 50% across aging, confirming the physical mechanism behind time-domain observer divergence. (3) *Thermal Robustness*: Under extreme low temperatures (0°C), internal impedance surges by ∼ 2.5×, increasing EKF RMSE tenfold, whereas the discontinuous control surface of the Sliding Mode Observer provides thermal invariance (0.929%–0.941% RMSE from 0°C to 45°C). (4) *Initialization and Convergence*: EKF and AEKF achieve rapid settling within 7.0 s and 6.0 s under severe −40% initial SOC uncertainty, whereas data-driven BiLSTM-Attn requires 30.0 s to clear its temporal sliding window. (5) *Flat-Plateau Observability*: On $LiFePO_4$ chemistry where $dV_{ocv}/d\text{SOC} \approx 0.08$ V, standard EKF errors surge by +1648%, while DEKF with dynamic hysteresis bounds RMSE to 0.870%, proving electrical observers track LFP cells without mechanical strain sensors. (6) *Sensor Fault Tolerance*: Under +0.25 A current sensor bias drift, open-loop Coulomb counting experiences linear runaway drift (12.399%), whereas closed-loop Kalman filters and SMO successfully arrest divergence, with SMO delivering lowest terminal drift (1.024%). (7) *Series Pack Inconsistency*: In an 8S series pack with manufacturing parameter variance ($\sigma_Q = 3.0\%, \sigma_R = 10.0\%$), a lumped macro-cell observer incurs a dangerous 1.744% RMSE and 2.302% peak error on the weakest cell, whereas distributed DEKF confines weakest-cell RMSE to 0.337%, guaranteeing pack safety margins. (8) *Embedded Pareto Optimality of PIDKF*: Cycle-accurate profiling on ARM Cortex-M4 and Cortex-M7 microcontrollers reveals that the proposed hybrid Physics-Informed Dual Kalman Filter (PIDKF) achieves superior tracking accuracy (0.414% RMSE on degraded NMC, 0.149% on degraded LFP at 85% SOH) with only 1,540 FLOPs and 14.20 μs execution latency on Cortex-M4, defining the empirical Pareto efficiency frontier for next-generation automotive BMS.

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A State-Space Matrices, Observability Gramians, and Analytic Jacobians

For the 3-state NMC and 4-state LFP models, the discrete state transition matrices $\mathbf{A}_k$ and observation Jacobians $\mathbf{H}_k$ are:

$$\mathbf{A}_k^{\text{NMC}} = \text{diag}\left(1, e^{-\Delta t/\tau_1}, e^{-\Delta t/\tau_2}\right), \tag{A1}$$

$$\mathbf{A}_k^{\text{LFP}} = \text{diag}\left(1, e^{-\Delta t/\tau_1}, e^{-\Delta t/\tau_2}, e^{-\beta|\Delta z_k|}\right), \tag{A2}$$

$$\mathbf{H}_k^{\text{NMC}} = \begin{bmatrix} V'_{\text{ocv}}(\widehat{\text{SOC}}_k^-) & -1 & -1 \end{bmatrix}, \tag{A3}$$

$$\mathbf{H}_k^{\text{LFP}} = \begin{bmatrix} V'_{\text{ocv}}(\widehat{\text{SOC}}_k^-) & -1 & -1 & -1 \end{bmatrix}, \tag{A4}$$

with $a_i = e^{-\Delta t/\tau_i}$ ($i \in \{1,2\}$) and $a_h = e^{-\beta|\Delta z_k|}$. The input vectors are $\mathbf{B}_k^{\text{NMC}} = [B_{1,k}, B_{2,k}, B_{3,k}]^T$ via Eq. (6), $\mathbf{B}_k^{\text{LFP}} = [\mathbf{B}_k^{\text{NMC}}; 0]$, and nonlinear excitation vector $\mathbf{g}_k = [0, 0, 0, (1 - a_h)M(z_k)\,\text{sgn}(I_k)]^T$. In DEKF, the Fréchet parameter sensitivity matrix $\mathbf{C}_{\theta,k} = \frac{dV_k}{d\theta}\Big|_{\hat{\theta}_k^-} \in \mathbb{R}^{1\times 2}$ is given by:

$$\mathbf{C}_{\theta,k} = \begin{bmatrix} -I_k & -\frac{\eta \Delta t I_{k-1}}{3600} V'_{\text{ocv}}(\widehat{\text{SOC}}_k^+) \end{bmatrix}. \tag{A5}$$

For the 3-state NMC model, the 3-step observability matrix is defined by:

$$O_{3,k} = \begin{bmatrix} \mathbf{H}_k \\ \mathbf{H}_k\mathbf{A}_k \\ \mathbf{H}_k\mathbf{A}_k^2 \end{bmatrix} = \begin{bmatrix} V'_{\text{ocv}}(z_k) & -1 & -1 \\ V'_{\text{ocv}}(z_k) & -a_1 & -a_2 \\ V'_{\text{ocv}}(z_k) & -a_1^2 & -a_2^2 \end{bmatrix}. \tag{A6}$$

Evaluating the determinant of $O_{3,k}$ via column operations yields the structured Vandermonde determinant:

$$\det(O_{3,k}) = V'_{\text{ocv}}(z_k)(1 - a_1)(1 - a_2)(a_1 - a_2). \tag{A7}$$

Because the electrochemical time constants are distinct ($\tau_1 = 24.0$ s $\neq \tau_2 = 112.5$ s $\implies a_1 \neq a_2$), sampling interval $\Delta t = 1.0$ s > 0 ensures $a_i \neq 1$, and polynomial calibration guarantees $V'_{\text{ocv}}(z) \geq \alpha > 0$ strictly, $\det(O_{3,k}) \neq 0$, proving uniform complete observability with condition number $\kappa(O_3) = \|O_3\|_2\|O_3^{-1}\|_2 \approx 3.2 \times 10^4$. For the 4-state LFP formulation, the augmented observability matrix determinant expands to:

$$|\det(O_{4,k})| = V'_{\text{ocv}}(z_k)(1-a_h)|a_1-a_2|\prod_{i=1}^{2}(1-a_i)|a_i-a_h|. \tag{A8}$$

Full rank $\text{rank}(O_{4,k}) = 4$ is strictly guaranteed under Persistent Excitation ($I_k \neq 0 \implies a_h < 1$).

B Lyapunov Stability, Finite-Time Reaching, and Error Boundedness of SMO

Define the scalar sliding manifold $S(t) = V_{\text{meas}}(t) - \hat{V}_t(t)$ with boundary layer continuous saturation $\text{sat}(S/\mu) = \max(-1, \min(1, S/\mu))$. The time derivative of $S(t)$ satisfies:

$$\dot{S}(t) = \Delta\dot{f}(t) - \gamma(\hat{z})\,\text{sat}\left(\frac{S}{\mu}\right), \tag{B1}$$

where $|\Delta\dot{f}| \leq \Phi < \infty$ bounds unmodeled dynamic disturbances, and the effective observer injection gain is defined as:

$$\gamma(\hat{z}) \equiv \mathbf{HL} = l_1 V'_{\text{ocv}}(\hat{z}) - l_2 - l_3. \tag{B2}$$

Since $V'_{\text{ocv}}(\hat{z}) \geq \alpha > 0$ for all physical states, choosing feedback gain $l_1 > (l_2 + l_3 + \Phi + \eta_s)/\alpha$ ensures $\gamma(\hat{z}) \geq \gamma_{\min} \geq \Phi + \eta_s > 0$ with strictly positive margin $\eta_s > 0$.

Consider the candidate Lyapunov function $V(S) = \frac{1}{2}S^2$.

1. Finite-Time Reaching Phase ($|S| > \mu$): Outside the boundary layer, $\text{sat}(S/\mu) = \text{sgn}(S)$. Evaluating the Lyapunov derivative yields:

$$\dot{V} = S\dot{S} = S\Delta\dot{f} - \gamma(\hat{z})|S| \leq -(\gamma_{\min} - \Phi)|S| \leq -\eta_s|S|. \tag{B3}$$

Substituting $|S| = \sqrt{2V}$ establishes the differential inequality:

$$\frac{dV}{dt} \leq -\eta_s\sqrt{2}\sqrt{V} \implies \frac{dV}{\sqrt{V}} \leq -\sqrt{2}\eta_s\,dt. \tag{B4}$$

Integrating from $t = 0$ to t_{reach} proves that the trajectory reaches the boundary layer $\mathcal{B}_\mu \equiv \{S \mid |S| \le \mu\}$ in finite time bounded by:

$$t_{\text{reach}} \le \frac{|S(0)| - \mu}{\eta_s} < \infty. \tag{B5}$$

2. Boundary Invariance Phase ($|S| \le \mu$): Inside $\mathcal{B}_\mu$, $\text{sat}(S/\mu) = S/\mu$, reducing error dynamics to $\dot{S} = -(\gamma/\mu)S + \Delta\dot{f}$. The Lyapunov derivative satisfies:

$$\dot{V} = -\frac{\gamma(\hat{z})}{\mu}S^2 + S\Delta\dot{f} \le -\frac{\gamma_{\min}}{\mu}S^2 + \Phi|S|. \tag{B6}$$

Therefore, $\dot{V} < 0$ holds strictly whenever $|S| > \mu\Phi/\gamma_{\min}$. Because $\gamma_{\min} > \Phi + \eta_s > \Phi$, the ratio $\Phi/\gamma_{\min} < 1$, which proves that the compact set:

$$\Omega_{\text{ult}} \equiv \left\{ S \;\middle|\; |S| \le \frac{\mu\Phi}{\gamma_{\min}} < \mu \right\} \subset \mathcal{B}_\mu \tag{B7}$$

is positively invariant, guaranteeing asymptotic chattering elimination.

3. Input-to-State Stability (ISS) of Polarization States: Within the invariant boundary layer Ω_{ult}, the continuous equivalent control signal satisfies $|u_{\text{eq}}| \le \Phi/\gamma_{\min}$. Polarization overpotential error dynamics $\dot{e}_{U_i} = -\frac{1}{\tau_i}e_{U_i} - l_{i+1}u_{\text{eq}} + d_{U_i}$ ($i \in \{1, 2\}$) are strictly exponentially Input-to-State Stable (ISS) with uniform asymptotic bound:

$$|e_{U_i}(t)| \le \tau_i \left(|l_{i+1}| \frac{\Phi}{\gamma_{\min}} + D_{U_i} \right) \equiv \bar{e}_{U_i} < \infty. \tag{B8}$$

4. Uniform Ultimate Boundedness (UUB) of State Error: By the Mean Value Theorem, thermodynamic voltage discrepancy satisfies $V_{\text{ocv}}(z) - V_{\text{ocv}}(\hat{z}) = V'_{\text{ocv}}(\xi)e_z$ for some $\xi \in [\min(z, \hat{z}), \max(z, \hat{z})]$. Consequently, the state estimation error is bounded by:

$$|e_z(t)| \le \frac{|S(t)| + |e_{U_1}(t)| + |e_{U_2}(t)| + |v(t)|}{V'_{\text{ocv}}(\xi)} \le \frac{\mu + \bar{e}_{U_1} + \bar{e}_{U_2} + V_{\text{noise}}}{\alpha}. \tag{B9}$$

Since $V'_{\text{ocv}}(\xi) \ge \alpha > 0$ strictly across $[0, 1]$, the denominator is strictly positive and bounded away from zero, mathematically proving that the complete estimation error vector $\|\mathbf{e}_x(t)\|$ is Uniformly Ultimately Bounded (UUB).

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